Medusa's Snakes

After the success of your latest research project in mythical DNA, you have gained the attention of a most diabolical creature: Medusa.

Medusa has snakes instead of hair. Each of her snakes' DNA is represented by an uppercase string of letters. Each letter is one of `S`, `N`, `A`, `K` or `E`.

Your extensive research shows that a snake's *venom level* depends on its DNA. A snake has venom level x if its DNA:

- has exactly $5x$ letters
- begins with x copies of the letter `S`
- then has x copies of the letter `N`
- then has x copies of the letter `A`
- then has x copies of the letter `K`
- ends with x copies of the letter `E`.

For example, a snake with venom level 1 has DNA `SNAKE`, while a snake that has venom level 3 has DNA `SSSNNNAAAKKKEEE`.

If a snake's DNA does not fit the format described above, it has a venom level of 0.

Medusa would like your help making her snakes venomous, by **deleting zero or more letters** from their DNA.

Given a snake's DNA, can you work out the maximum venom level this snake could have?

Input

The first line contains the integer N: the number of letters in the snake's DNA. The second line contains a string of N uppercase letters, representing the snake's DNA. Each letter is one of `S`, `N`, `A`, `K` or `E`.

Output

Your program should output a single integer: the maximum venom level the snake could have, after you delete some (possibly none) of the letters from its DNA.

Sample Input 1

```
17
KSEESNANNAAKNKESE
```

Sample Output 1

```
2
```

Sample Input 2

```
22
SSSSNNNAAAAKKKKEEEEEEE
```

Sample Output 2

```
3
```

Sample Input 3

```
15
SNAKESNAKESNAKE
```

Sample Output 3

```
1
```

Sample Input 4

```
6
KANSAS
```

Sample Output 4

```
0
```

Explanation

The letters that are deleted in each case are underlined below:

- Sample Input 1: KSEESNANNAAKNKESE → SSNNAAKKEE
- Sample Input 2: SSSSNNNAAAAAKKKKEEEEEEE→ SSSNNNAAAKKKEEE
- Sample Input 3: SNAKESNAKESNAKE → SNAKE
- Sample Input 4: No matter which letters you delete, the snake will always have venom level 0, so the answer is 0.

Subtasks & Constraints

For all cases, $5 \leq N \leq 100\,000$. Additionally:

- For Subtask 1 (15 marks), all `S` come before all `N`, which come before all `A`, which come before all `K`, which come before all `E`. There will be at least one of each letter. Sample Input 2 is an example of a case that could be in this subtask.
- For Subtask 2 (15 marks), the DNA sequence consists of `SNAKE` repeated some number of times. Sample Input 3 is an example of a case that could be in this subtask.
- For Subtask 3 (30 marks), $N \leq 10$.
- For Subtask 4 (20 marks), $N \leq 1000$.
- For Subtask 5 (20 marks), no further constraints apply.